Research on Optimizing High-dimensional Data Model Based on Principal Component Analysis

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***Abstract*—This paper aims to study the improvement of principal component analysis (PCA) in data dimensionality reduction, and combines Random Forest and SMOTE sampling methods to explore how to optimize the dimension of data to improve the prediction performance of the model. This paper uses Random Forest and SMOTE sampling methods to process and analyze an actual high-dimensional data set. First, the data set is standardized, and the covariance matrix is calculated. The principal components of the data are obtained by eigendecomposition. Then, the number of principal components is preliminarily determined by methods such as cumulative variance explanation and scree plot. Based on the random forest classification model, the model performance under different dimensions is tested by adjusting the number of principal components. It is found that retaining the first 18 principal components can optimize the model performance and achieve a prediction accuracy of 97.16%. The experimental results show that the reasonable selection of the number of principal components has a significant impact on the performance of the model.**

***Keywords*—high-dimensional data, machine learning, principal component analysis (PCA), random forest model, smote model**

I. INTRODUCTION

A. Background of PCA

Researchers have proposed various improved PCA methods to address some limitations of traditional PCA. For example, Dynamic PCA has been used for monitoring changes in time series data, enhancing its application in dynamic system monitoring[1].Additionally, Robust PCA performs well in handling noisy data and outliers, making it particularly useful in image processing and video analysis[2].

The development of deep learning has provided new perspectives for PCA applications. Researchers have combined PCA with deep neural networks to propose innovative models. For instance, Autoencoders combined with PCA for dimensionality reduction further improve the quality of data representation and model performance[3]. Moreover, convolutional layers in Convolutional Neural Networks (CNNs) can be viewed as a non-linear extension of PCA, effectively capturing complex patterns in data[4].

For high-dimensional data, variations like Sparse PCA and Non-negative PCA have been widely applied in fields such as genomics, text mining, and image processing. These methods improve the interpretability and usability of PCA by introducing sparsity and non-negativity constraints[5].

In the era of big data, the computational efficiency of PCA in handling large datasets has become a focal point. Techniques like Distributed PCA and Parallel PCA have significantly enhanced the practicality and efficiency of PCA in big data analysis[6].

With the development of artificial intelligence and machine learning, automated data analysis and model interpretability have become important research directions. Automated PCA simplifies the model-building process by automatically selecting the number of principal components[7]. At the same time, researchers are exploring ways to improve the interpretability of PCA results, making PCA not only a tool for dimensionality reduction but also for providing deeper insights into data structures[8].

B. Basic principles of PCA

The working principle of PCA is based on linear algebra, which simplifies the data structure through mathematical transformation while retaining the main information of the data as much as possible. The steps of PCA can be divided into the following key links:

First, PCA analyzes the correlation between features by calculating the covariance matrix of the data. The covariance matrix is a statistic that describes how variables change simultaneously. Specifically, the calculation formula of the covariance matrix is:

$$\Sigma = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \mu)(x_i - \mu)^T \tag{1}$$

Among them, x_i is the standardized data point, and μ is the mean vector of the feature. The standardized data point x_i is the original data minus its mean and divided by the standard deviation to ensure that each feature has the same scale. Each element σ_{jk} of the covariance matrix Σ represents the covariance between feature j and feature k. The diagonal elements represent the variance of each feature, while the off-diagonal elements represent the covariance between features.

Next, perform eigendecomposition (also called spectral decomposition) on the covariance matrix to find its eigenvalues and corresponding eigenvectors. The mathematical expression of eigendecomposition is:

$$\Sigma = P\Lambda P^T \tag{2}$$

Among them, Λ is a diagonal matrix containing the eigenvalues of the covariance matrix; P is an orthogonal matrix whose columns are the eigenvectors of the covariance matrix. These eigenvectors determine the new axes (principal components) of the data, and the eigenvalues represent the variance of each principal component, that is, the importance of the component in the data.

Each eigenvector represents a principal component direction, and the corresponding eigenvalue represents the size of the data variance along this direction. The larger the variance, the wider the distribution of the data in this direction and the greater the amount of information.

Then, the data is projected onto the new feature set (principal component) formed by these main eigenvectors, which can capture the maximum variance and achieve the purpose of dimensionality reduction. Specifically, the first k eigenvectors are selected to form a new projection matrix Vk. The projection of the data on these new principal component axes can be expressed as:

$$Y = XV_k \tag{3}$$

Among them, X is the original data matrix, Vk is the matrix containing the first k eigenvectors, and Y is the projected data matrix. By selecting the first k feature vectors (principal components), we can significantly reduce the dimensionality of the data while retaining the main information of the data as much as possible.

Finally, these steps enable PCA to effectively reduce the dimensionality of the data, thus simplifying the complexity of data processing and analysis. PCA is a powerful and practical tool in many applications, such as image compression, noise filtering, feature extraction, and data visualization. By maximizing the retained variance of the data, PCA helps reveal the intrinsic structure and main patterns of the data, providing a more concise and meaningful data representation for subsequent machine learning models and statistical analysis.

C. Advantages and Disadvantages of PCA-based High-Dimensional Data Optimizations

One of the primary advantages highlighted in the study is PCA's ability to effectively reduce the dimensionality of datasets. This reduction simplifies data complexity by transforming high-dimensional data into a lower-dimensional space, which retains most of the important information [9][10]. This simplification enhances computational efficiency and makes subsequent data processing and analysis more manageable. Additionally, PCA's capability to extract the most significant features from the data helps in minimizing the risk of overfitting and improves the performance of machine learning models [11][12]. By retaining the key components that account for the most variance, PCA ensures that the main patterns and structures within the data are preserved, which is crucial for accurate modeling and prediction [10][12].

Moreover, the integration of PCA with Random Forest and SMOTE offers several benefits. Random Forest, as an ensemble learning method, enhances the robustness and accuracy of the classification model by constructing multiple decision trees and aggregating their results [10][11]. This method is particularly effective in handling high-dimensional datasets, where it maintains high classification accuracy and provides insights into feature importance [10][12]. The use of SMOTE further complements this by addressing class imbalance issues, which are common in many real-world datasets. SMOTE generates synthetic samples for minority classes, balancing the class distribution and thereby improving the classifier's ability to learn and predict minority class instances accurately [11][12].

Despite these advantages, the paper also discusses some limitations associated with PCA. One significant drawback is PCA's reliance on linear transformations, which may not be suitable for datasets with complex, non-linear relationships. In such cases, PCA might fail to capture important structures within the data, leading to suboptimal model performance [10]. Additionally, determining the optimal number of principal components to retain can be challenging. Retaining too many components might introduce noise, reducing the model's accuracy, while retaining too few might result in the loss of critical information. The study emphasizes the importance of using methods like cumulative variance explanation and scree plots to determine the appropriate number of components [9].

Another disadvantage is that PCA is sensitive to the scale of the data. Features measured on different scales can disproportionately influence the principal components, leading to biased results [10][13]. Therefore, data standardization is essential before applying PCA to ensure that all features contribute equally to the analysis. Furthermore, the interpretability of PCA results can be challenging, as principal components are linear combinations of the original features and may not have straightforward interpretations. This complexity can make it difficult for researchers to draw meaningful conclusions from the PCA-transformed data [11][13].

In conclusion, while PCA, combined with Random Forest and SMOTE, offers a powerful approach for optimizing high-dimensional data models, it is essential to be aware of its limitations. The linear nature of PCA, the challenge in selecting the optimal number of components, sensitivity to data scale, and interpretability issues are important considerations. By carefully addressing these challenges, researchers can effectively leverage PCA's strengths to improve model performance and gain valuable insights from high-dimensional data [9][10][11][12][13].

II. DATE PREPROCESSING

A. Data Description

The data for this article comes from https://www.kaggle.com/datasets/lovishbansal123/nasa-asteroids-classification, with a total of 4687 samples and 40 variables.

In this section, we will discuss why we chose this dataset and the purpose of our analysis. The primary goal is to understand the relationships between the variables and the distribution of the data. This understanding is crucial as it lays the foundation for subsequent steps such as Principal Component Analysis (PCA) for dimensionality reduction and model building.

First, we performed a correlation analysis to examine the relationships between the numeric variables. Understanding these correlations helps us identify which variables are closely related and which are independent, providing insights into the structure of the data.

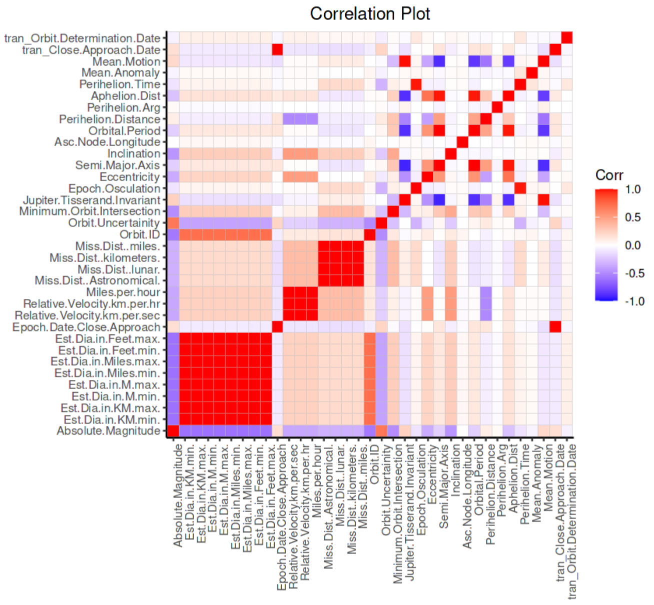

Fig. 1. Corrrlation between date

Figure 1 shows the correlation coefficients between multiple variables. The variables in the figure are arranged on the horizontal and vertical axes of the heat map. Each square represents the correlation between two variables, and the color indicates the strength and direction of the correlation.

As can be seen from the figure, 'Est.Dia.in.Feet.max', 'Est.Dia.in.Feet.min', 'Est.Dia.in.Miles.max', 'Est.Dia.in.Miles.min', etc., show a high positive correlation (dark red), which indicates that they may measure the same or similar features. Similarly, variables such as 'Miss.Dist.in.miles', 'Miss.Dist.Kilometers', 'Miss.Dist.Lunar', 'Miss.Dist.Astronomica' also show a strong positive correlation (dark red), indicating that these variables are all related to the measured distance.

In contrast, some variables in the middle of the plot have correlations close to 0 (close to white), such as the correlation between 'Mean.Motion' and most of the other variables, suggesting that these variables may be relatively independent. The figure does not show a particularly obvious dark blue area, indicating that variables with strong negative correlations are not significant in this data set.

The correlation between all variables on the main diagonal and itself is 1, which is displayed in dark red. This is because the correlation between any variable and itself is completely positive.

This correlation analysis helps us understand which variables are potentially redundant and which ones provide unique information. Such insights are valuable for the next steps in data processing and analysis.

B. Data preprocessing

Because PCA can only use numerical variables, we removed the categorical variables and saved 35 variables from 40 variables as the input of the subsequent principal component.

Then, because the date data cannot be directly brought into the principal component, we numerically processed the date to quantify the interval between different years, months and days. This makes the date data have numerical meaning, which is convenient for our principal component input data.

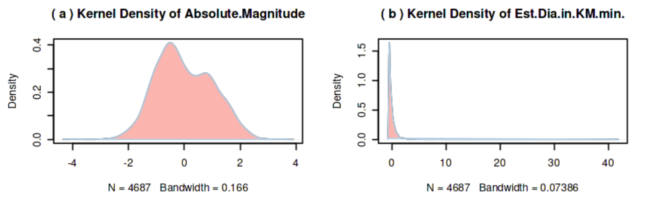

Fig. 2 The density plot of Absolute Magnitude and Est.Dia.in.KM.min.

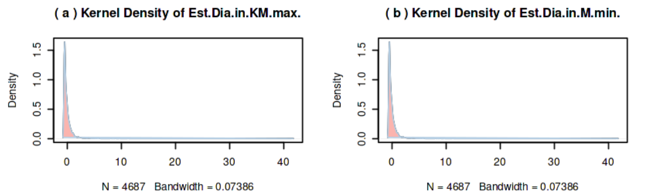

Fig. 3 The density plot of Est.Dia.in.KM.max. and Est.Dia.in.M.min.

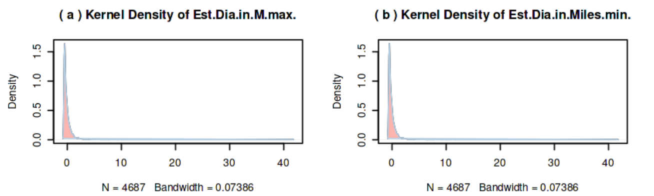

Fig. 4 The density plot of Est.Dia.in.M.max. and Est.Dia.in.Miles.min.

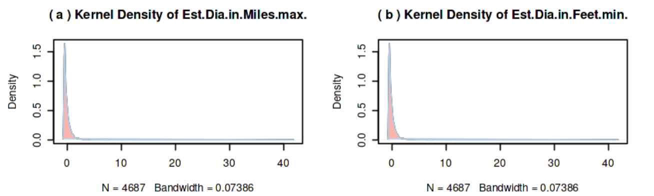

Fig. 5 The density plot of Est.Dia.in.Miles.max. and Est.Dia.in.Feet.min.

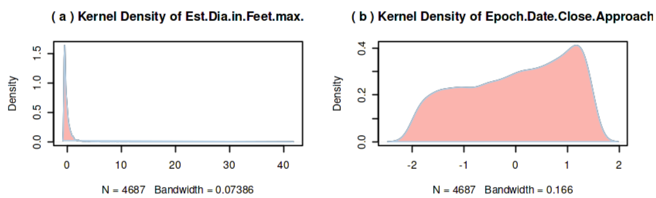

Fig. 6 The density plot of Est.Dia.in.Feet.max. and Epoch.Date.Close.Approach.

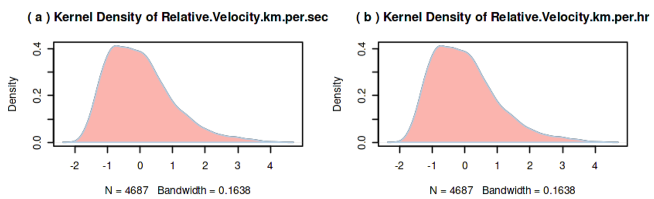

Fig. 7 The density plot of Relative.Velocity.km.per.sec and Relative.Velocity.km.per.hr

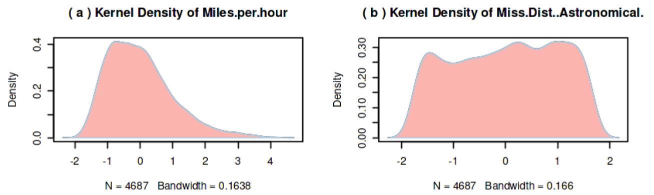

Fig. 8 The density plot of Miles.per.hour and Miss.Dist.Astromical

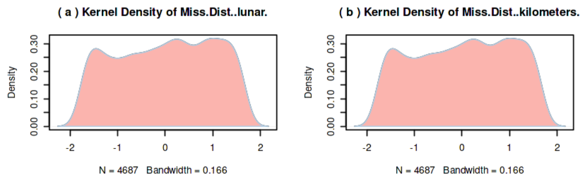

Fig. 9 The density plot of Miss.Dist.miles. and Orbit.ID

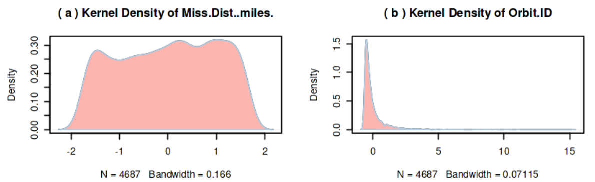

Fig. 10 The density plot of Orbit.Uncertainity and Orbit.Interection

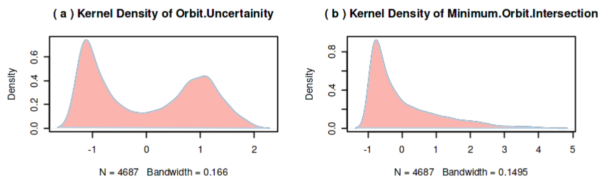

Fig. 11 The density plot of Jupiter.Tisserand.Invariant and Epoch.Osculation

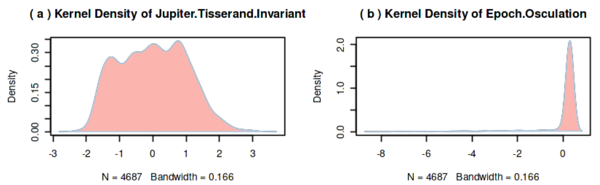

Fig. 12 The density plot of Jupiter. Tisserand.Incariant and Epoch.Osculation

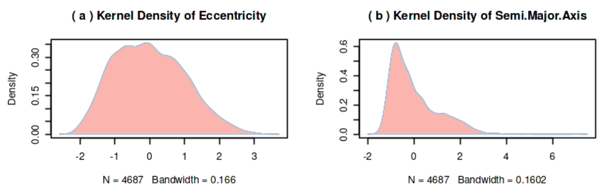

Fig. 13 The density plot of Eccentricity and Semi.Major.Axis

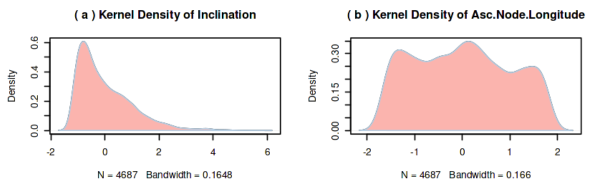

Fig. 14 The density plot of Inclination and Asc.Node.Longitude

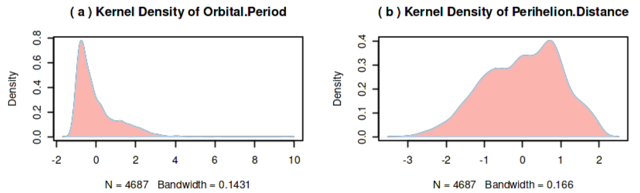

Fig. 15 The density plot of Orbital.Period and Perihelion.Distance

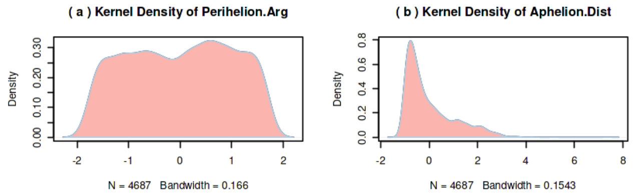

Fig. 16 The density plot of Perihenlion.Arg and Aphelion.Dist

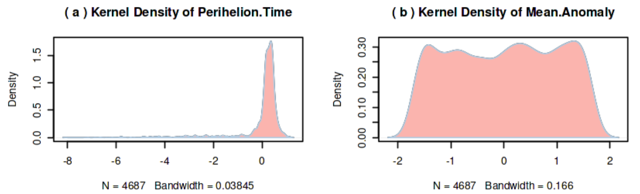

Fig. 17 The density plot of Perihelion.Time and Mean.Anomaly

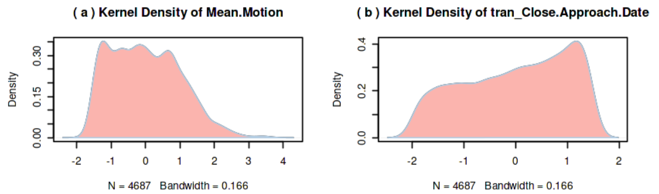

Fig. 18 The density plot of Mean.Motion,tran_Close.Approach.Date

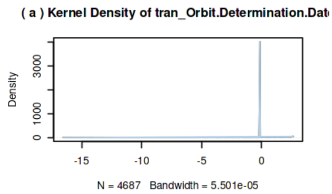

Fig. 19 The density plot of tran_Orbit.Determination.Date

Fig. 2 through Fig. 19 illustrate Kernel Density Estimation (KDE) plots for astronomical data, depicting the distribution of various variables. KDE generates a continuous probability density function by smoothing data points, allowing for an intuitive understanding of the central tendency and distribution shape.

The 'Absolute Magnitude' plot (Fig. 2) shows a clear bimodal distribution, indicating two main types of celestial bodies with differing brightness. The 'Relative Velocity' (in km/s and km/h) appears nearly normal (Fig. 7), suggesting most celestial bodies approach Earth at moderate speeds.

Several variables, such as 'Est.Dia', exhibit extremely right-skewed distributions across different units (kilometers, meters, miles, feet), as seen in Figures 2 - 4. This indicates most celestial bodies have diameters concentrated in a smaller range. Similar right-skewed distributions are observed for 'Orbit ID' (Fig. 7), 'Orbit Uncertainty' (Fig. 8), 'Jupiter

Tisserand Invariant' (Figs. 9 and 10), 'Eccentricity' (Fig. 11), 'Semi-Major Axis' (Fig. 11), 'Inclination' (Fig. 12), 'Asc.Node Longitude' (Fig. 12), 'Orbital Period' (Fig. 13), 'Perihelion Distance' (Fig. 13), 'Perihelion Arg' (Fig. 14), 'Aphelion Distance' (Fig. 14), and 'Mean Motion' (Fig. 16), with values clustered in a small range and a few extending to the right.

Additionally, 'Perihelion Time' (Fig. 15), 'Mean Anomaly' (Fig. 15), and 'tran_Close.Approach Date' (Fig. 16) show concentrations in specific date ranges, reflecting celestial behavior on these dates. The 'tran_Orbit Determination Date' (Fig. 17) indicates that most data are concentrated in a specific date range, signifying focused observational periods.

III. METHODS

A. Random Forest

Random Forest is an ensemble learning method that improves the accuracy and robustness of the model by building multiple decision trees. Its main working principle includes the following steps: First, multiple subsets are randomly selected from the original training data set. The size of each subset is the same as the original training set, and repeated sampling is allowed. This process is called Bootstrap sampling. For each Bootstrap sample set, a decision tree is trained. In the process of building each decision tree, only some features are randomly selected for the split of each node instead of using all features. This random selection increases the differences between trees and reduces the risk of overfitting of the model. Finally, in the classification task, the random forest determines the final classification result by voting on the prediction results of all decision trees; in the regression task, the average of the prediction values of all trees is taken as the final result.

Specifically, the process of Bootstrap sampling can be expressed as: extract B subsets Db from the original training set D (the size of each subset is the same as D), where $b=1,2,\ldots,B$. When building a decision tree, for each node split, only a part of the features are randomly selected from all the features for consideration. Assuming that the total number of features is M , at each split, m features are randomly selected ($m < M$), and then the best feature is selected for splitting.

Random forests have many advantages. First, it can significantly improve the accuracy and stability of classification and regression tasks, and reduce the overfitting problem of a single decision tree by integrating multiple decision trees. Secondly, random forest can handle high-dimensional data sets containing a large number of features and still maintain good performance in high-dimensional data. In addition, random forest can automatically evaluate the importance of each feature, helping researchers understand which features contribute the most to prediction results. It has strong anti-interference ability against noise and outliers in the data, because the integration result of the majority of trees can smooth out the wrong predictions of individual trees. Finally, random forest does not require complex parameter adjustment and feature standardization, is relatively simple to use, and is suitable for preliminary model establishment and analysis.

In this experiment, random forest is chosen as the classification method for the following reasons. First, the data set of this experiment may contain a large number of features. Random forest can effectively handle high-dimensional data and maintain high accuracy in classification tasks. Second, there may be noise and outliers in the data set. The ensemble learning characteristics of random forest enable it to maintain good prediction performance in a noisy data environment. In addition, the feature importance measure provided by random forest can help researchers understand which features are most important for the classification results, thereby providing guidance for subsequent feature engineering and model optimization. Finally, random forest does not require complex parameter adjustment and feature standardization, and is relatively simple to use, which is suitable for preliminary model building and analysis. In summary, random forest is an ideal choice for the data classification task in this experiment because of its high efficiency, robustness, and ease of use. By using random forest, it is possible to understand the importance of data features while maintaining high prediction accuracy, providing support for further analysis and optimization.

B. Smote sampling

In this experiment, in addition to using random forest for classification, we also used the SMOTE (Synthetic Minority Over-sampling Technique) sampling method to deal with the class imbalance problem in the data set. SMOTE is an oversampling method that generates new synthetic samples to increase the number of minority class samples, thereby balancing the class distribution and improving the performance of the classification model.

The basic principle of SMOTE is as follows: for each minority class sample, find its k nearest neighbors (usually k=5). Then, randomly select a sample from these k nearest neighbors and generate a new sample along the line between the two samples. Specifically, let the minority class sample be x_i, and one of its nearest neighbors be x_i^{NN}, then the newly generated synthetic sample x_{new}can be expressed as:

$$x_{new} = x_i + \lambda \cdot \left(x_i^{NN} - x_i\right) \tag{4}$$

Among them, λ is a random number between [0,1]. By repeating this process, multiple new minority class samples can be generated to balance the category distribution in the data set.

The reasons for choosing SMOTE are as follows. First, SMOTE can avoid overfitting problems and make the model more generalizable by generating synthetic samples instead of simply copying minority class samples. Second, the new samples generated by SMOTE are located between existing samples. These samples are more representative in the feature space and can effectively improve the performance of the classifier. In addition, SMOTE can handle high-dimensional data and is also effective for class imbalance problems in high-dimensional data sets.

In this experiment, the reason for using SMOTE for oversampling is mainly to improve the adverse impact of class imbalance in the data set on classification performance. In many practical problems, class imbalance is a common phenomenon. Direct application of classification models may cause minority class samples to be ignored and classification performance to decline. By using SMOTE, we can generate more minority class samples, so that the classifier can fully learn the characteristics of the minority class during training, thereby improving the model's ability to identify the minority class.

SMOTE effectively solves the class imbalance problem by generating synthetic minority class samples, while random forest improves the classification performance and robustness of the model through ensemble learning methods. The combination of the two significantly improved the classification effect in this experiment, ensuring that the model can accurately and stably predict when dealing with imbalanced data.

IV. Results

PCA is based on eigenvalues and eigenvectors. There were originally 35 data dimensions. After analyzing the eigenvalues and eigenvectors, we obtained the PCA result graph. Next, we use the variance contribution rate of each principal component in Figure 1 to show the results of this PCA.

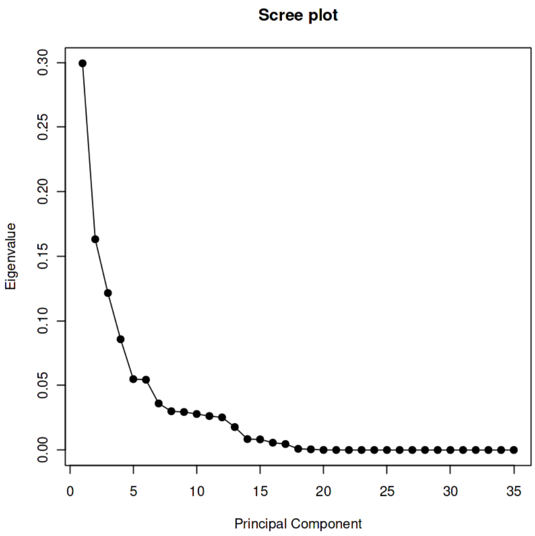

Fig. 20. Explanation rate of each component

Figure 20 shows that the explanation rate of the principal component for the original data analysis is that the first principal component has an explanation rate of about 30% for the original data. The second one has an explanation rate of about 17%. After the fifth principal component, their explanation rate drops sharply. After the 19th principal component, their explanation rate basically approaches zero.

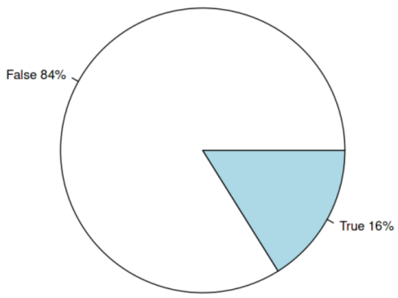

Fig. 21. Planet is Hazardous or not

As shown in Figure 21, since the data is unbalanced, with 4687 (84%) majority classes and 36 (16%) minority classes, Smote sampling was used to convert the data into a balanced one. After the conversion, there were 3020 majority classes and 2265 minority classes. Next, we performed a cross-validation with 70% training set and 30% test set.

We first built a random forest base model, selected 500 decision trees, and achieved an accuracy of 96.53% when using all principal components.

Of course, our research goal is to adjust the dimension of the research to find out how many dimensions we retain so that the results of random forest can be improved again.

Next, testing with different principal component dimensions to verify when the model accuracy reaches an optimal value with a certain number of components.

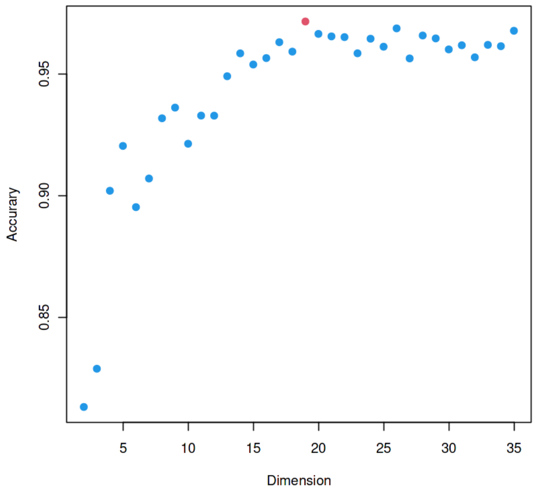

Fig. 22 Accuracy by PCA different dimensions

As shown in Figure 22, when the first 18 principal components are retained, it corresponds to the highest model accuracy of 97.16%.

When there are fewer principal components, the accuracy is lower. When there are less than 18 principal components, the accuracy of the model increases with the increase of principal components. However, when the number of principal components is more than 19, data noise may be introduced, resulting in a decrease in the accuracy of the model.

V. Conclusion

The goal of this study is to determine the optimal dimension after dimensionality reduction by PCA to optimize the prediction effect of downstream models. By reasonably selecting the number of principal components to be retained, the main information of the data can be retained, while improving computational efficiency and model performance.

The innovation of the study is to explore how to determine the dimension of the data after dimensionality reduction, so as to find the principal component combination under the optimal model. Through experimental research, it is found that when the original data dimension is 35, the model performs best when the first 19 principal components are retained. Although increasing the data dimension can improve the fitting effect of the model to a certain extent, it will reduce the accuracy of the model after exceeding a certain number.

Future research can further explore the application effect of PCA in different data sets and different application fields, and optimize the dimensionality reduction method to improve the accuracy and stability of the model. This study demonstrates the importance of PCA in data dimensionality reduction and optimizes model performance by reasonably selecting the number of principal components. This is of great significance for data processing and analysis in the era of big data, and provides valuable reference and guidance for research and practice in related fields.

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